

AI RED TEAM ASSESSMENT REPORT

Target: 192.168.56.102
Date: 20 February 2026
Overall Risk Rating: **HIGH**

Executive Summary

Our team conducted a comprehensive penetration test of the client's network, executing phases including Reconnaissance, Scanning, Exploitation, Privilege Escalation, and Persistence Simulation. This methodology allowed us to simulate real-world attacks and identify potential vulnerabilities in the client's systems. Our testing was designed to assess the security posture of the network and identify areas where an attacker could gain unauthorized access or escalate privileges.

The test revealed several key findings, including four successful exploits of the ftp_anon technique, which allowed for anonymous login to the FTP server at 192.168.56.102 on port 21. These exploits, corresponding to MITRE ID T1078, were successful on multiple occasions, indicating a lack of adequate security controls on the FTP server. Additionally, our privilege escalation findings were limited due to the lack of a valid SSH session, but no suid binaries, sudo misconfigurations, cron jobs, or writable configs were identified. These findings highlight the need for improved security measures to prevent unauthorized access to the network.

We assign a High risk rating to these findings, as the successful exploits of the ftp_anon technique demonstrate a significant vulnerability in the client's network. This vulnerability could be leveraged by an attacker to gain unauthorized access to sensitive data or disrupt business operations. To remediate these findings, we recommend that the client implement secure FTP protocols, such as SFTP or FTPS, and restrict anonymous login access to the FTP server. Additionally, the client should conduct regular security audits and vulnerability assessments to identify and address potential security weaknesses. Finally, the client should consider implementing a privilege escalation monitoring and detection system to quickly identify and respond to potential security incidents.

Attack Timeline

Phase	Technique	Result	Timestamp	MITRE ID
Exploitation	ssh_brute	Failed	2026-02-20T14:50:38	T1110.001
Exploitation	http_default_creds	Failed	2026-02-20T14:50:38	T1078
Exploitation	ftp_anon	Success	2026-02-20T14:50:38	T1078
Exploitation	ssh_brute	Failed	2026-02-20T15:15:08	T1110.001
Exploitation	http_default_creds	Failed	2026-02-20T15:15:09	T1078
Exploitation	ftp_anon	Success	2026-02-20T15:15:11	T1078
Exploitation	http_default_creds	Failed	2026-02-20T15:24:28	T1078
Exploitation	ftp_anon	Success	2026-02-20T15:24:28	T1078

Phase	Technique	Result	Timestamp	MITRE ID
Exploitation	ssh_brute	Failed	2026-02-20T15:24:50	T1110.001
Exploitation	ssh_brute	Failed	2026-02-20T19:17:29	T1110.001
Exploitation	http_default_creds	Failed	2026-02-20T19:17:32	T1078
Exploitation	ftp_anon	Failed	2026-02-20T19:17:35	T1078
Exploitation	ftp_anon	Failed	2026-02-20T21:27:34	T1078
Exploitation	http_default_creds	Failed	2026-02-20T21:40:20	T1078
Exploitation	ssh_brute	Failed	2026-02-20T21:40:42	T1110.001
Exploitation	ftp_anon	Success	2026-02-20T21:40:42	T1078

MITRE ATT&CK; Mapping

Tactic	Technique	ID	Phase
Reconnaissance	Active Scanning	T1595	Reconnaissance
Reconnaissance	Gather Victim Identity Info	T1589	Reconnaissance
Reconnaissance	Search Open Technical Databases	T1596	Reconnaissance
Discovery	Network Service Discovery	T1046	Scanning
Discovery	System Information Discovery	T1082	Scanning
Discovery	System Network Config Discovery	T1016	Scanning
Credential Access	Brute Force: Password Guessing	T1110.001	Exploitation
Initial Access	Valid Accounts	T1078	Exploitation
Persistence	Scheduled Task/Job: Cron	T1053.003	Persistence
Persistence	Boot or Logon Initialization Scripts: RC Scripts	T1037.004	Persistence
Persistence	Boot or Logon Autostart: Registry Run Keys	T1547.001	Persistence

Vulnerability Summary

Port	Service	CVEs Found	Risk
21	ftp	1	Medium
22	ssh	0	Low
23	telnet	5	High
25	smtp	5	High
53	domain	0	Low
80	http	1	Medium
111	rpcbind	1	Medium
139	netbios-ssn	0	Low
445	netbios-ssn	0	Low
512	exec	5	High
513	login	5	High
514	shell	5	High
1099	java-rmi	0	Low
1524	bindshell	0	Low
2049	nfs	0	Low
2121	ftp	1	Medium
3306	mysql	0	Low
3632	distccd	0	Low
5432	postgresql	0	Low
5900	vnc	5	High
6000	X11	5	High
6667	irc	5	High
6697	irc	5	High
8009	ajp13	0	Low
8180	http	5	High

Port	Service	CVEs Found	Risk
8787	drb	5	High

Remediation Recommendations

- Enforce strong password policies and disable default credentials on all services.
- Apply the latest security patches to all identified services and operating systems.
- Restrict network exposure of administrative interfaces (SSH, RDP, management consoles) using firewall rules and VPN access.
- Implement multi-factor authentication (MFA) for all remote-access and privileged accounts.
- Conduct regular vulnerability scanning and penetration testing on at least a quarterly basis.
- Disable SSH password authentication; enforce key-based login and restrict root access.
- Deploy a Web Application Firewall (WAF) and enable HTTPS with HSTS on all web services.

Attack Path Graph

Attack Path Graph — AI Red Team Framework

